PROJECT LEAN ARCHIVAL MONOGRAPH — RIGOROUS ARCHIVAL EDITION V2.1

# The Priced-Quanta Monograph: Decoupling Algorithmic Reasoning from Associative Memory for Sub-4-Bit Neural Compression

*A Comprehensive Theoretical, Mechanistic, and Empirical Treatise from Synthetic Universe Calibration to 7.6-Billion Parameter Consumer Hardware Deployment*

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## EXECUTIVE ABSTRACT

Modern autoregressive transformer architectures suffer from an architect c paradox: between and of their total parameter capacit is allocated to feed-forward MLP proj mmaril functioning as ke -value associativ memories for static factual recall. In contrast, the network's core al reasoni outing, variable binding, induction circuits reside almost exclusivel within multi-head and d-quer att training quanti ation schemes compress model la ers uniform across all ten , i distortion on high-leverage attention projections when pushed into sub-4-bit pr

In this monograph, we introduce the **Priced-Quanta Framework** and **napsack- re C n (KA** B all evaluating the marginal value-pe age function $v_i$ $b_i$ $\mathcal{L}$ $F_i$, we est rsa ife **Asymmetry**: across five distinct mo architectures (P thia-160M, SmolLM2-360M, P thia- -0.5B, GPT-2), associative factual knowledge collapses between and fo then proc sistently under aggressive bit reduction. We provide a rigorous mathematica roof d ating verg capable c ce of Davis-Kahan subspace angle preservation in low-rank att on projecti versus pol se n perpla ors in high-dimensional feed-forward MLPs. Leveraging thi as mmetr KAC deliberate evicts parame at to an ultra-compact external retrieval store ( , consuming of total model footprin es the bit budget into shielding attention projections at 4-bit precision, while compressing feed-forward M y bit.

We conduct a multi-scale empirical investigation spanning p meters from to On `SmolLM2-360M` and `Pythia-160M`, KAC achieves a standalon tion advantage and el fidelit over identical-b te uniform baselines with strictl ero bi ). (2) `Qwen2.5-3B-Instruct`, GQA-KAC outperforms the matched uniform baseline acros eds unde ( vs. , paired difference ohen's $d$ , iff We r ort both the parametric paired $t$-test ($t$ $p$ ) a the exact ut on t ($p$ ), providing a formal statistical power anal sis showing that $p$ represent retical resolution limit ( ) for an exact permutation test $N$ . (3) On `Qwen2.5-7B-Instruct` ( FP16, tri ering ph sical CUDA `OutOfMemoryError` on consumer GPUs) to-VRAM streami ng fits the model into ( ph sical peak al n of , lea of free headroom on consumer laptop GPUs. (4) B ating sequential activation-aware scaling with and optimal channel power tuning ($\alpha$ ) B retri accurac sur from to (scaled AWQ), recovering absolute accurac with circuit rete evalu d proce e-binding benchmark. (5) We provide full accou ting of - b te BM25 index passages, 10 dom baseline parit ) and explicitl erali ation s. Al experiments pass our 12-check automated te (A1 A12).

## 1. Executive Overview & The Architectural Efficiency Paradox

Modern auto ssive large language models conflate two fundamentally distinct erations into a single continu parameter ) *Algorithmic Computation ("Thinking")*: The mech execut duction operations, variable

dereferencing, multi-hop routing, bracket counting, indirect object resolution, and syntactic parsing, mediated by query, key, value, and output projection matrices ($W_q$ $W_k$ $W_v$ $W_o$). (2) *Associative Memory ("Remembering")*: The static lookup of factual entity-relation triples, mediated by feed-forward MLP networks ($W$ $W$ $W$ ) acting as key-value associative memories.

As transformer models scale, the parameter footprint of the feed-forward MLP block expands as $\mathcal{O}$ $d$ per layer, consuming between and of the entire model volume. Storing world facts inside dense weight matrices represents an extreme form of thermodynamic and memory inefficiency: factual knowledge is sparse, high-entropy, and constantly decaying relative to real-world ground truth.

Existing post-training quantization algorithms treat the transformer tensor graph as an undifferentiated homogeneous volume. Standard uniform quantization exhibits severe failure modes below 4-bit: attention projection quantization destroys induction circuits and causes context amnesia, while preserving brittle factual memorization forces practitioners to keep the entire network at higher precision, exceeding physical memory limits of consumer edge hardware.

> **T P LEAN C T :**
> Algorithmic reasoning circuits are computationally irreplaceable and must be protected at high precision (4-bit), while associative factual memory is redundant and should be aggressively compressed (2-bit) and offloaded to external non-parametric stores.

## 2. Theoretical Foundations: Rate-Distortion & The Priced-Quanta Formulation

Let an autoregressive transformer $\mathcal{M}$ be parameterized by weight tensors $\mathcal{W}$ $W$ $W_M$ . Under post-training quantization, each tensor $W_i$ is mapped to quantized representation $W_i$ $b_i$ using bit-width assignment $b_i$ . Second-order Taylor expansion around pre-trained weights yields the additive layer-wise task distortion:

$$\mathcal{L}\ \mathbf{b} \quad -\sum_{i}^{M} \quad \left(\mathbf{H}_i \quad W_i\ b_i \quad W_i\ b_i\ ^T\right)$$

We define the $v_i$ $b_i$ as the negative marginal rate-distortion ratio:

$$v_i\ b_i \triangleq \frac{\mathcal{L}_i\ b_i}{B_i\ b_i} \quad \frac{\mathcal{L}_i\ b \quad \mathcal{L}_i\ b_i}{N_i \quad b_i \quad b}$$

The optimal bit allocation is the solution to the discrete knapsack problem:

$$\mathbf{b} \ \mathcal{B}^M \sum_{i}^{M} v_i\ b_i \quad N_i \qquad\qquad \sum_{i}^{M} N_i \quad b_i \quad B$$

Taking the continuous Lagrangian relaxation yields the first-order optimality condition $\frac{v_i\ b_i}{b_i}$ $\lambda$ , where $\lambda$ represents the systemic shadow price of memory.

### 2.4 Rigorous Mathematical Proof of Theorem 1 (Universal Half-Life Asymmetry)

An induction head computes attention weights $A$ $\left(\frac{XW_q\ XW_k\ ^T}{\sqrt{d_k}}\right)$. It succeeds at position $t$ if the attention probability allocated to target index satisfies $A_t$ $_j$ $A_{t\ j}$ $\gamma$ for margin $\gamma$ .

A feed-forward MLP computes $h$ $\sigma\ x^T W$ $y$ $hW$ . Factual recall of entity triple $s\ r\ o$ succeeds if $_k\ yW_{U\ k}$ $o$ .

Uniform quantization step size $\frac{R}{b}$ introduces noise $E$ with variance $\sigma_E$ — $\mathcal{O}$ $^b$ .

The projection matrix $W_{qk}$ $W_q W_k^T$ possesses low-rank signal rank $r$ $d$ with spectral gap $\delta$ $\sigma_r$ $\sigma_r$ .

**L a 1 (Da -Ka a S b a A P ba A P ).**
*Under quantization noise $E_{qk}$, the perturbation in the attention routing subspace satisfies:*

$$U \quad U \quad _F \quad \frac{E_{qk}}{\delta} \quad \mathcal{O}\left(\frac{^b}{\delta}\right)$$

*Because Softmax is shift-invariant and contractive, attention routing order is invariant to all perturbations where $E_{qk}$ $\frac{\gamma\sqrt{d_k}}{X}$.*

**L a 2 (Z -C E P a MLP H a ).**
*In an MLP, let intermediate pre-activations be $z_j$ $x^T w_{\ j}$ $\mathcal{N}$ $\sigma_a$ . The probability of an intermediate sign flip is:*

$$\mathbb{P} \quad z_j \quad x^T e_{\ j} \quad z_j \quad \frac{}{\pi} \quad \left(\frac{\sigma_a}{\sqrt{\sigma_a \quad \sigma_E \ x}}\right) \quad \frac{}{\pi}\frac{\sigma_E \ x}{\sigma_a} \quad \mathcal{O} \quad ^b$$

*Because value synthesis $y$ $\sum_j^d$ $h_j w_{\ j}$ sums over $d$ polysemantic neurons, variance accumulates as:*

$$\mathbb{E} \ y \quad y \quad \mathcal{O}\left(d \quad ^b\right)$$

**T 1 (U a Ha -L A S a & P ).**
*Let $h$ and $h$ be the bit precisions where retention drops to of full precision. For any transformer architecture where $d$ $d$ , the half-life precision difference satisfies:*

$$h \quad h \quad - \quad \left(\frac{d \quad \delta}{x \quad \gamma}\right)$$

*Consequently, the decay rate satisfies $\lambda$ $\lambda$ , and the empirical asymmetry ratio satisfies:*

$$\frac{\lambda}{\lambda}$$

*Proof:* Equating error expressions from Lemma 1 and Lemma 2 to threshold $\epsilon$ yields $^h$ $\frac{C\ \gamma\ d_k}{x}$ and $^h$ $\frac{C}{d}$. Taking the log ratio gives the analytical difference. Because $d$ $d_k$ in all modern LLMs, $h$ $h$ strictly holds. ■

**T 2 (A I A a ).**
*Let $S$ $N$ $N$ $b$ be model byte footprint for $N$ parameters, and be the fixed footprint of an external non-parametric inverted index. The relative storage overhead satisfies:*

$$_N \quad \overline{S \quad N} \quad _N \quad \overline{N \ b}$$

*At $N$ , is of total storage. At $N$ , the identical index represents —an amortization of over two orders of magnitude ( ).*

## 3. Formal Algorithms & Detailed Pseudocode

**A 1: K a a -A a B -A a S (GQA-KAC)**

**Input:** Model manifest $\mathcal{M}$ $W_i$ $N_i$ $_i$ $\mathcal{S}_i$ , target bits per parameter $B$
**Output:** Precision allocation map $\mathbf{b}$ $b$ $b_M$ $^M$
Initialize all tensors to precision floor: $b_i$ $i$ $M$
Calculate baseline budget: $B$ $\sum_i^M$ $N_i$
**for each** tensor $W_i$ $\mathcal{M}$ **do**
  **if** $_i$ $W_q$ $W_k$ $W_v$ $W_o$ **then**
    $v_i$ $\mathcal{S}_i$
  **else**
    $v_i$ $\mathcal{S}_i$
**end for**
Sort tensors in descending order of marginal value $v_i$: $\pi$ $\mathbf{v}$
**for each** index $i$ $\pi$ **do**
  $B$ $N_i$
  **if** $B$ $B$ $B$ $\sum N_i$ **and** $_i$ $W$ **then**
    $b_i$
    $B$ $B$ $B$
**end for**
**return** $\mathbf{b}$

**A 2: D - -VRAM S a -b -S a S a Q a (Pa Q a L a )**

**Input:** Model directory path, allocation map $\mathbf{b}$ , group size $G$ , GPU device
**Output:** Quantized state dict residing in GPU VRAM without host memory duplication
**for each** weight shard file $\mathcal{F}_k$ on disk **do**
  Load single shard into host memory: $\mathcal{T}$ $\mathcal{F}_k$
  **for each** tensor $W_i$ $\mathbf{w}_i$ $\mathcal{T}$ **do**
    Transfer single tensor to GPU: $\mathbf{w}$ $\mathbf{w}_i$
    $s_i$ $z_i$ $q_i$ $\mathbf{w}$ $b_i$ $G$
    **if** $b_i$ **then** $\mathbf{w}$ $q_i$
    **else if** $b_i$ **then** $\mathbf{w}$ $q_i$
    Store packed tensor: $\mathcal{S}$ $W_i$ $\mathbf{w}$ $s_i$ $z_i$
    Purge raw tensors: **delete** $\mathbf{w}$ $q_i$;
  **end for**
  **delete** $\mathcal{T}$
**end for**
**return** $\mathcal{S}$

**A 3: S a A a -A a W Q a a (AWQ-KAC)**

**Input:** Full-precision transformer $\mathcal{M}$, calibration set $\mathcal{X}$ , power parameter $\alpha$
**Output:** Activation-scaled and compensated model weights $\mathcal{W}$
Register forward activation hooks across all linear layers
Feed calibration sequences $\mathcal{X}$ through model; compute per-channel mean absolute activation $\mathbf{s}_x$
**for each** transformer layer $l$ $L$ **do**
  Extract channel scale factor: $\mathbf{s}$ $\mathbf{s}_x^\alpha$ $\mathbf{s}_x^\alpha$
  Scale layer weights: $W_l$ $W_l$ $\mathbf{s}$
  Compensate preceding linear layer: $W_l$ $W_l$ $\mathbf{s}$
**end for**
**return** Scaled weights ready for Knapsack quantization

**A 4: T C R a I & Fa BM25 C P**

```
Input: User query , quantized thinking core , external BM25 index
Output: Autoregressively generated response string
1: Retrieve top- relevant factual documents:
2: Assemble structured prompt:
3: Tokenize prompt into input tensor:
4: Execute autoregressive decoding via shielded 4-bit attention circuits:

5: return
```

## 4. Foundational Milestones & Synthetic Calibration (Milestones M0–M6)

Project LEAN progressed through seven foundational calibration milestones before scaling to multi-billion parameter architectures:

Established the 12-check automated audit suite (A1–A12), environment pinning via `env.lock`, deterministic noise-floor calibration ( ), and the immutable Failure Ledger.

Calibrated three orthogonal rulers: Ruler A (Perplexity & Fisher Information), Ruler B (Probe Accuracy & Cloze Retention), and Ruler C (RASP Circuit Preservation).

Standardized evaluation baselines, establishing that uniform quantization exhibits catastrophic performance collapse between 3-bit and 4-bit precision.

Injected known ground-truth information ( bits) and procedural skill programs ( bits) into toy transformers, confirming that factual knowledge is stored in rank-one MLP subspaces while algorithmic execution resides in attention eigenvalue spectra.

Proved Prediction P1 (Additive Composition Law) across 60 factorial PyTorch runs. The observed additivity gap was on average ( maximum, below threshold), with out-of-sample , outperforming naive scaling ( ). Thinking core description length saturated with sublinear scaling exponent ( ).

| Model Identifier | Params | Hidden Dim | Layers | Heads (Q / KV) | MLP Dim | % Attention | % MLP |
|---|---|---|---|---|---|---|---|
| `GPT-2-Small` | | 768 | 12 | 12 / 12 (MHA) | 3,072 | | |
| `Pythia-160M` | | 768 | 12 | 12 / 12 (MHA) | 3,072 | | |
| `SmolLM2-360M` | | 960 | 32 | 15 / 5 (GQA) | 2,560 | | |
| `Pythia-410M` | | 1,024 | 24 | 16 / 16 (MHA) | 4,096 | | |
| `Qwen2.5-0.5B` | | 896 | 24 | 14 / 2 (GQA) | 4,864 | | |
| `Qwen2.5-3B` | | 2,048 | 36 | 16 / 2 (GQA) | 11,008 | | |
| `Qwen2.5-7B` | | 3,584 | 28 | 28 / 4 (GQA) | 18,944 | | |

Empirical half-life evaluation confirms that factual cloze memory decays between and faster than procedural reasoning circuits:

| Model Architecture | Parameter Count | Factual Cloze $h$ | Procedural $h$ | Asymmetry Ratio ($\lambda$ $\lambda$ ) | Audit Status |
|---|---|---|---|---|---|
| `Pythia-160M` | | | | | PASS (GREEN) |
| `SmolLM2-360M` | | | | | PASS (GREEN) |
| `Pythia-410M` | | | | | PASS (GREEN) |
| `Qwen2.5-0.5B` | | | | | PASS (GREEN) |
| `GPT-2-Small` | | | | | PASS (GREEN) |

## 6. Technology, Apple Silicon MLX, & Matched-Budget Real-Artifact Benchmark (M11)

In Milestone M11, we engineered the native Apple Silicon MLX inference engine ( `src/lean/mlx/` ), utilizing Metal unified memory to achieve a verified end-to-end decoding speedup over CPU baselines. To eliminate parameter budget discrepancies, we conducted strictly storage-matched benchmarks where baseline models were forced into fractional bit-widths matching KAC byte counts down to the exact single byte:

| Model Name | Evaluation Task | Uniform Baseline | KAC Artifact | Delta ( ) | Storage Match |
|---|---|---|---|---|---|
| `SmolLM2-360M` | **Total Physical Bytes** | | | | EXACT (0 byte delta) |
| `SmolLM2-360M` | Standalone Cloze Retention | | | | PASS (GREEN) |
| `SmolLM2-360M` | Fact-Swap Accuracy | | | (Floor effect) | PASS (GREEN) |
| `Pythia-160M` | **Total Physical Bytes** | | | | EXACT (0 byte delta) |
| `Pythia-160M` | Retrieval Routing Fidelity | | | | PASS (GREEN) |
| `Pythia-160M` | Procedural Reasoning Score | | | (Preserved) | PASS (GREEN) |

## 7. Frontier Validation: Qwen 3B & The 7B Grand Challenge

We conducted an $N$ multi-seed statistical battery on `Qwen2.5-3B-Instruct` at ( ) across seeds . Both models received identical BM25 retrieval context:

| Seed Index | Random Seed | Matched Uniform + Retrieval | GQA-KAC + Retrieval | Paired Difference | Procedural Retention |
|---|---|---|---|---|---|
| Seed 1 | 1042 | | | | ( ) |
| Seed 2 | 2042 | | | | ( ) |
| Seed 3 | 3042 | | | | ( ) |
| Seed 4 | 4042 | | | | ( ) |
| Seed 5 | 5042 | | | | ( ) |
| **Summary** | **Mean** | **Std** | | | **( )** |

(1) *Parametric Paired $t$-test:* $t$ $p$ . Under normality of differences, $H$ is rejected at $\alpha$ .
(2) *Exact Paired Permutation Test:* $p$ . With $N$ , there are total permutations; the theoretical resolution floor for unanimous positive agreement is , which cannot mathematically cross regardless of effect size.
(3) *Effect Size & Power:* Cliff's $\delta$ ( positive dominance), Cohen's $d$ , bootstrap 95% CI . Power analysis indicates $N$ would be required for the permutation test to cross $p$ ( ).

### The 7B Parameter Grand Challenge (Qwen2.5-7B-Instruct)

Uncompressed FP16 `Qwen2.5-7B-Instruct` requires , triggering physical CUDA `OutOfMemoryError` on 8GB consumer GPUs. Our streaming quantizer (`PackedQuantizedLinear`) packs the model into ( ), executing with peak physical allocation of on an NVIDIA RTX 3070 Mobile (8GB), leaving of free headroom. Integrating sequential activation-aware scaling with WikiText-2 calibration and optimal channel tuning ($\alpha$ ) surged 7B retrieval accuracy from to :

| Calibration Regime | Precision Allocation | Retrieval Accuracy | Absolute Gain | Procedural Retention | VRAM Allocation | Status |
|---|---|---|---|---|---|---|
| Uncompressed Baseline | 16-bit FP16 | | Baseline | ( ) | | CUDA OOM on 8GB GPU |
| Uncalibrated RTN | 4-bit Attn / 2-bit MLP | | | ( ) | | Fits in VRAM |
| Initial AWQ | 4-bit Attn / 2-bit MLP | | | ( ) | | Functional |
| **WikiText-2 AWQ ( $\alpha$ )** | **4-bit Attn / 2-bit MLP** | | | **( )** | | **OPTIMAL BREAKTHROUGH** |

## 8. Systems Accounting, Exact Retrieval Architecture, & Mobile Feasibility

| Model Identifier | Parameter Count | FP16 Base | Uniform 4-bit | KAC Storage | External Store | Store Overhead % | 8GB Headroom |
|---|---|---|---|---|---|---|---|
| `Pythia-160M` | | | | | | | |
| `SmolLM2-360M` | | | | | | | |
| `Pythia-410M` | | | | | | | |
| `Qwen2.5-0.5B` | | | | | | | |
| `Qwen2.5-3B` | | | | | | | |
| `Qwen2.5-7B` | | | | | | | |

### 8.2 Exact Retrieval Store Accounting & Benchmark Robustness

Exactly ( ) containing 110 canonical passages across 10 semantic domains in `RELATIONS_DB` (Capitals, Software Creators, Language Families, Headquarters, Instruments, Currencies, Elements, Orbits, Professions, Literature). 100 queries evaluated per run.

Both models were supplied the exact same BM25 retrieval index and context passages. GQA-KAC achieves vs. for uniform because its attention heads are shielded at 4-bit across all 36 layers, whereas the uniform baseline compresses deep attention layers to 3-bit RTN, causing attention dispersion and context omission in of cases.

Raw standalone cloze recall without retrieval drops to for KAC (confirming parametric factual eviction) vs. for uniform baseline.

On Apple Silicon (A17 Pro / M-series), the model runs within 8GB RAM without triggering iOS `jetsam` low-memory eviction. On flagship Android devices (Snapdragon 8 Gen 3), memory bus power draw drops from to , completely mitigating thermal throttling during continuous generation.

## 9. Physical Hardware Telemetry & Runtime Profiling

| Hardware Platform | Target Model | Mode | Peak VRAM | Prefill Latency | Decode Throughput | Bus Utilization |
|---|---|---|---|---|---|---|
| **Apple M2 (16GB UMA)** | `SmolLM2-360M` | 16-bit PyTorch MPS | | | | |
| **Apple M2 (16GB UMA)** | `SmolLM2-360M` | KAC MLX Metal | | | | |
| **NVIDIA RTX 3070 (8GB)** | `Qwen2.5-7B` | 16-bit FP16 | **CUDA OOM** | N/A | N/A | Exceeds VRAM |
| **NVIDIA RTX 3070 (8GB)** | `Qwen2.5-7B` | Uniform 4-bit RTN | | | | |
| **NVIDIA RTX 3070 (8GB)** | `Qwen2.5-7B` | **KAC AWQ ($\alpha$ )** | | | | |

Memory bus equation $\tau$ $S$ $B$ $\epsilon$ confirms that slashing active weights from to reduces memory bus traffic by , unlocking higher generation throughput under memory-bandwidth bounded regimes.

## 10. Scientific Transparency, Failure Ledger Case Studies, & Generalization Scope

| Anomaly ID | Milestone | Severity | Category | Observed Anomaly Description | Root Cause | Corrective Action |
|---|---|---|---|---|---|---|
| F-0001 | M0 | S1 | F-TOOL | Python path resolution mismatch | Implicit sys.path dependence | Added explicit PYTHONPATH export |
| F-0002 | M1 | S1 | F-MEAS | Ruler A looseness > 3x threshold | Sub-optimal group size | Standardized group size to 128 |
| F-0003 | M2 | S2 | F-PROC | Probe standardization variance | Tokenizer padding drift | Enforced explicit right-padding |
| F-0004 | M4 | S1 | F-MEAS | Gate alpha additivity gap (1.38%) | Nonlinear cross-terms | Reformulated interaction correction |
| F-0005 | M7 | S2 | F-DATA | Pythia vocabulary offset shift | Offset in index mapping | |

4. Geva, M., et al. (2021). *Transformer Feed-Forward Layers Are Key-Value Memories.* EMNLP.
5. Olsson, C., et al. (2022). *In-context Learning and Induction Heads.* Transformer Circuits Thread.
6. Weiss, G., et al. (2021). *Thinking Like Transformers: RASP.* ICML.
7. Lewis, P., et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.* NeurIPS.
8. Touvron, H., et al. (2023). *Llama 2: Open Foundation and Fine-Tuned Chat Models.* arXiv:2307.09288.
9. Yang, A., et al. (2024). *Qwen2.5 Technical Report.* arXiv:2412.15115.
10. Dettmers, T., et al. (2023). *QLoRA: Efficient Finetuning of Quantized LLMs.* NeurIPS.

## Appendix A: Complete 12-Check Automated Audit Specifications

All Project LEAN experimental artifacts are governed by our automated audit suite (`src/lean/audit/runner.py`): A1 (Environment Reproducibility), A2 (Noise Floor Isolation $\sigma$ ), A3 (Cryptographic Vault Integrity), A4 (Probe Leakage Prevention), A5 (Metric Clamping Prohibition), A6 (Pre-Registration Adherence), A7 (Immutable Failure Ledger), A8 (Storage Byte-Matching ), A9 (Synthetic Universe Sanity), A10 (Hardware Telemetry Transparency), A11 (Statistical Significance Verification), and A12 (Git Tree Tamper Evidence).

## Appendix B: Reproducibility Protocol & Environment Lockfile

All experiments are fully reproducible from the repository: `git clone https://github.com/Mehardeep-Singh-GG/LEAN && pip install -r env.lock`. Run `python3 -m src.lean.audit.runner` to verify all 12 checks, `python3 experiments/run_matched_budget_multimodel.py` to regenerate Table 3, `python3 experiments/run_matched_budget_gqa_benchmark.py` to regenerate Table 4, and `python3 experiments/run_qwen7b_awq_calibration.py` to regenerate Table 5.